

Initial state

1 2 3
4 - 6
7 8 5

$$f(n) = g(n) + h(n)$$

$$= 1 + 2$$

$$= 3$$

1 - 3

4 2 6

7 8 5

1 2 3

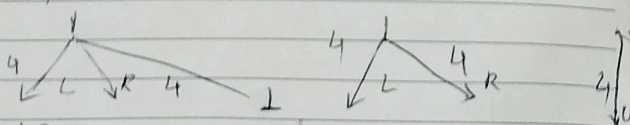
4 8 6

7 - 5

1 2 3

- 4 6

7 8 5



- 1 3 1 3 - closed 1 2 3 1 2 3
4 2 6 4 2 6 4 8 6 4 8 6
7 8 5 7 8 5 - 7 5 7 5 -

Optimal path

right → down → left ⇒ 3

Goal

1 2 3
4 5 6
7 8 -

$h(n)$ = misplace
tile

(blank excluded)

$g(n)$ = each step
(up, down, left, right) = 1

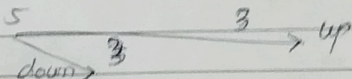
3

right

1 2 3

4 6 -

7 8 5



- 2 3 1 2 3 7 8 - 7 8 5
1 4 6 7 4 6 - 8 5 2 4 / L
7 8 5 - 8 5 2 ~~up~~ 1 - 2

1 2 3

4 5 6

7 8 -

4 6 3

7 8 5